

ANURAG PARSI

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4X Certified Data Scientist

Open To: Data Scientist | Data Analyst | Data Engineer | Gen AI | Onsite/Hybrid/Remote

I am: Data Expert(LLM, Fine-tuning, Data Prediction, Data Visualization, Data Modelling, Statistical modeling, Data Collection)

Recent Hands-On: All ML Algorithms (Classification, Regression, Ensemble Techniques), Deep Learning (ANN, CNN RNN), NLP (NLTK, TensorFlow, Keras, PyTorch), LLM Models, Prompt Engineer, Gen AI.

Hands-on-Domains: Telecom, Sales, Finance, Healthcare, Insurance, Retail.

[LinkedIn Id](#)

[Know Me More](#)



Professional Summary:

I am an experienced Data Scientist with over 5 years of expertise in developing and deploying machine learning, NLP, and generative AI solutions, including large language models (LLMs). Proficient in leveraging cloud platforms like AWS and Databricks, I specialize in building scalable, end-to-end AI systems for industries such as healthcare and finance. With strong expertise in Python and frameworks like TensorFlow and PyTorch, I excel at driving innovation and delivering data-driven strategies. Passionate about applying advanced AI technologies, I thrive on solving complex challenges and aligning solutions with business goals.

Technical Skills

Programming Languages: Python, MySQL, R Language, Scala, PySpark, Java, Java scripts.

DBMS/No SQL databases: SQL Server, MySQL, PL/SQL, Oracle, COBOL, Druid, T/SQL, PostgreSQL, Cassandra, Dynamo DB.

AWS Services: Sagemaker, Athena, EC2, EMR, S3, Lambda, Glue, CloudWatch, RDS, Redshift.

Azure Services: Azure Data Lake, Azure Data Factory, Azure SQL, Synapse Analytics, Azure SQL, Azure ML, Azure Kubernetes.

GCP Services: Vertex AI, Dataflow, Dataflow, BigQuery, Dataproc, AI Hub, Cloud Logging.

Python Packages: Scikit-Learn, NLTK, TensorFlow, PyTorch, Keras, saCpy, Pandas, NumPy, Seaborn, Matplotlib, Beautiful Soup

Statistics: Mean, Median, Mode, Data Distributions, Standard Deviation and Variance, Hypothesis Testing (p-values) and Test for significance- t-test, Chi-Square test, A/B Testing and test for normality, Bayesian Inference.

Machine Learning: Feature Selection Supervised and Unsupervised Learning, Reinforcement Learning, Correlation Analysis, Linear Regression, Multiple Linear Regression, Logistics Regression, Clustering, Classification, Decision Tree, Support Vector Machines (SVM), Naive Bayes, Ensemble Techniques (Random Forest, XGBoost, Gradient Boosting Machines (GBM)), K-Nearest Neighbors (KNN), Clustering, K-Means Clustering, DB-Scan Clustering, SAS, ARIMA.

Natural Language Processing (NLP) and Deep Learning: ANN, CNN, RNN, Hugging Face Transformers, PyTorch, TensorFlow, Keras, LSTMs, Transformers, Graph Neural Networks, ELMo, BERT, OpenAI GPT-4, BART.

Version Control/ETL: Airflow, Git, GitLab, GitHub, Jenkins, Docker, Kubernetes, Terraform, DevOps, MLOps, Mlflow, Data Version Control (DVC).

Visualization: Tableau, PowerBI, Excel, Streamlit, Matplotlib, Pandas, Ploty, Seaborn, D3.js, R Shiny.

Methodologies: Software Development Life Cycle (SDLC), Jira, Agile, Waterfall Model.

Soft Skills: Leadership, Mentoring, Communication skills, Presentation skills, Problem Solving, team oriented.

EXPERIENCE

Verizon

April 2024 – Present

Role: Senior Data Scientist

- Designed and deployed AI-driven predictive models to enhance **fiber optic expansion** and resource allocation, utilizing Python and Golang for scalable solutions.
- Developed **RESTful APIs** using **FastAPI** to facilitate seamless integration of **AI/ML** models into enterprise applications, ensuring efficient data exchange.
- Implemented advanced NLP techniques with **spaCy**, **Transformers**, and **NLTK** to analyze customer interactions, improve service recommendations, and optimize network operations.
- Applied vector search algorithms, leveraging **Azure Vector DB** and **FAISS** for high-performance data retrieval in fiber optic network planning and customer segmentation.
- Utilized Azure services, including OpenAI, Event Grid, and **Azure Kubernetes Service (K8S)**, for deploying scalable AI solutions and streamlining model management.
- Engineered predictive models (**Random Forest**, **Gradient Boosting**, **LSTMs**) and **fine-tuned BERT** for sentiment analysis on Azure ML, achieving a 20% boost in prediction accuracy, enabling more precise customer sentiment insights.
- Designed and maintained microservices architectures, ensuring cloud-based deployments were optimized for performance, security, and reliability.
- Worked extensively with **NoSQL** databases, including **MongoDB** and **vector-based** databases, to support high-throughput AI-powered analytics.
- Leveraged containerization (**Docker**) and orchestration (Kubernetes) to optimize **ML** model deployment, ensuring scalability and efficient resource utilization.
- Integrated AI solutions with enterprise collaboration tools, such as Microsoft Teams and Edge plugins, improving cross-functional communication and operational workflows.
- Applied problem-solving and analytical skills to develop and optimize fiber network expansion models, reducing deployment costs and enhancing efficiency.
- Maintained **agile development** practices, meeting tight deadlines while collaborating with cross-functional teams to drive innovation in AI-powered network solutions.
- Successfully managed multiple projects simultaneously within an **agile environment**, ensuring timely delivery of data-driven solutions

- Developed and optimized shipping APIs, improving logistics workflows within the fiber optic deployment ecosystem.
- Conducted **A/B** testing and statistical modeling to refine AI models, ensuring optimal accuracy and alignment with business goals.
- Designed and implemented scalable AI pipelines for real-time predictive analytics, enhancing operational decision-making
- Provided technical leadership on **AI/ML** projects, mentoring junior team members and fostering a culture of continuous learning.

Aadhyam Solutions Private Limited

Feb 2020 – March 2024

Role: Data Scientist

- **Collaborated with stakeholders** to understand business requirements, leading data analytics and modeling projects that increased success rates by 25%. Utilized advanced statistical methods, including **regression analysis, A/B testing, hypothesis testing, and PCA** to extract actionable insights, improving decision-making
- Expertise in integrating and analyzing **EHR(Electronic Health Records), claims (ICD, CPT,HCPCS), genomic (GWAS, biomarkers), imaging (DICOM, radiology AI),** wearable/IoT, pharmaceutical, and public healthcare data, with skills in **CDSS, interoperability (HL7, FHIR), and HIPAA** compliance.
- Developed algorithms for processing and interpreting large data sets to drive actionable insights.
- Applied advanced **statistical techniques** to analyze and interpret complex healthcare data sets.
- Leveraged research skills to improve existing data mining techniques, enhancing overall business intelligence
- Designed and implemented scalable data pipelines using **AWS Glue, Lambda, and Airflow**, achieving a 30% boost in data processing efficiency. Developed and integrated **APIs**, including **REST, FastAPI, and GraphQL**, streamlining data exchange and enhancing system interoperability.
- Built and deployed machine learning models such as **Logistic Regression, SVM, Random Forest, and XGBoost** on **AWS SageMaker**, addressing business-critical use cases like customer care predictions and resource forecasting with up to 25% accuracy improvement. Optimized imbalanced datasets with hyperparameter tuning for better **precision, recall, and F1 Scores**.
- Created interactive dashboards using **Tableau** and **Power BI**, improving real-time decision-making by 25%. Enhanced stakeholder engagement by 20% with data analysis and visualization in **Jupyter Notebooks** on **AWS EC2**, making complex insights accessible and actionable.
- Reduced operational costs by 20% with **AWS S3** and **RDS** for scalable data management. Streamlined workflows using **MLOps** and **Docker** for containerized model deployment. Automated **CI/CD** pipelines with **GitHub Actions**, accelerating delivery and enhancing project efficiency by 25% using **Agile** and **Scrum** practices.

CERTIFICATES

- [Microsoft Certified: Azure Data Scientist Associate](#)
- [Data Science Foundation](#)
- [Analytics Vidya: Introduction to Data Science](#)
- [Data Visualization with Power BI](#)

Education

Master of Science in Data Science at New England College | Henniker, NH

Aug 2022 - May 2024

Research Experience

> **Author:** Anurag Parsi -: [Publication_Link](#) **Feb 2022-May 2022**

Integrated Artificial Intelligence (**AI**) and Machine Learning (**ML**) algorithms to enhance the chatbot's conversational capabilities. Improving response accuracy by 25% through pattern matching and AI Markup Language (AIML) to enhance chatbot performance and provide more relevant responses. Collaborated with a team to expand the chatbot’s knowledge base with AI

PERSONAL PROJECTS:

- 1)House Price Prediction: [Demo_Video](#) | [Launch_Application: -](#)
- Challenge: Estimating property prices was difficult due to market variability and low user engagement.
- Solution: Developed an interactive house price prediction tool enabling users to choose **ML models** and **visualize pricing trends, feature correlations, and predictions**.
- 2)Search Engine with tools and Agents: [Demo_Video](#) | [Launch_Application: -](#)
- Problem: Researchers faced inefficiencies in retrieving specific information from large knowledge bases.
- Solution: Developed a **search engine** leveraging **LangChain, OpenAI-4,** and **LLM** capabilities, integrating tools like Wikipedia and arXiv for efficient data retrieval. Implemented prompt engineering, **retrieval-augmented generation (RAG),** fine-tuning, Similarity Search, VectorDBs, and custom agents to enhance context understanding and deliver accurate results.
- 3)Appicate Tracking System (ATS): [Demo_Video](#) | [Launch_Application: -](#)
- Problem: Talented Job seekers often failed to align their resumes with job descriptions, resulting in low visibility in ATS.
- Solution: Built an ATS tool using **LangChain** and **LLMs** to analyze resumes against job descriptions. Increased shortlisting rates by 40% through keyword alignment and actionable improvement suggestions, saving HR teams 30%